A COMPARATIVE STUDY OF HRTF MEASUREMENTS AND SIMULATIONS USING SONICOM DATA

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***Abstract*—Head-related transfer functions (HRTFs) are essential for binaural rendering and spatial audio applications. While measured HRTFs provide high accuracy, numerical simulations offer a flexible alternative for subject-specific modelling. In this work, head-only and head-and-torso transfer functions are numerically simulated for multiple human subjects taken from SONICOM, a publicly available dataset that provides, for each person, both acoustically measured transfer functions and a three-dimensional scan of the head and body. The analysis compares the proposed simulations with the official SONICOM simulated HRTFs using spectral error, log-spectral distortion (LSD), interaural level difference (ILD), and interaural time difference (ITD). Results are analysed over frequency bands and spatial regions to assess whether the addition of a torso improves the agreement with the measured data. The head-and-torso simulation shows physically meaningful improvements below 6 kHz and in low-elevation lateral directions. LSD analysis confirms an improvement of the lateral spectral agreement, while ITD provides only a secondary indication of improved lateral temporal behaviour and ILD does not improve globally.**

***Index Terms*—HRTF, Mesh2HRTF, SONICOM, binaural audio, torso simulation, ITD, ILD, LSD**

I. Introduction

Head-related transfer functions (HRTFs) describe the direction-dependent acoustic filtering between a sound source and the two ears of a listener. They are essential for binaural rendering, virtual reality, augmented reality, and headphone-based spatial audio. The spatial information encoded in HRTFs is strongly related to the listener's morphology: interaural time differences (ITDs) and interaural level differences (ILDs) mainly contribute to lateralisation, while direction-dependent spectral cues, largely shaped by the pinnae, head, shoulders, and torso, are crucial for elevation perception and front–back discrimination [1].

Individual HRTFs can be obtained through acoustic measurements, but this procedure requires specialised equipment, accurate microphone placement, and controlled acoustic environments. For this reason, numerical HRTF synthesis has become an important alternative. Boundary element method (BEM)-based simulation tools, such as Mesh2HRTF, allow listener-specific HRTFs to be computed from three-dimensional anatomical meshes, while mesh grading and fast multipole acceleration reduce the computational cost of full-band simulations [3], [4].

Large HRTF datasets are fundamental for validating such numerical approaches. Previous databases have combined measured and simulated HRTFs with anthropometric and mesh data, showing that cross-evaluation between measurements and simulations is a meaningful way to assess HRTF generation pipelines [5]. More recently, the SONICOM dataset has provided measured HRTFs, 3D scans, photogrammetry data, and simulated HRTFs for a large number of subjects, making it a suitable reference for reproducible research in personalised spatial audio [6]. The extended SONICOM release further includes synthetic HRTFs generated using Mesh2HRTF - code available on GitHub [1]- and introduces the Spatial Audio Metrics (SAM)[2] toolbox for numerical comparison of HRTF datasets [7].

Although head and pinna morphology are often the main focus of HRTF simulation studies, the torso and shoulders can also affect the HRTF through reflection, diffraction, and shadowing. Previous work has shown that shoulder reflections may introduce comb-filter structures, while torso obstruction can lead to high-frequency damping, especially for low-elevation and lateral directions [9]. However, the practical benefit of adding a torso to a subject-specific HRTF simulation is not always straightforward, because the torso geometry must also match the measured subject accurately.

The aim of this work is to assess how faithfully a subject-specific Mesh2HRTF pipeline can reproduce a measured HRTF and, in particular, whether adding a torso to the simulated geometry improves this agreement. To this end, four HRTF sets for the same subject are compared: the measured SONICOM HRTFs, the official SONICOM simulated HRTFs, used as a benchmark for the residual error inherent to any simulation, our head-only Mesh2HRTF simulation, and our head-and-torso Mesh2HRTF simulation. The head-only model first serves to validate the pipeline against the measurement, whereas the head-and-torso model isolates the acoustic contribution of the upper body. Agreement is quantified using signal-based and binaural metrics: the mean spectral error and log-spectral distortion (LSD) describe differences in HRTF magnitude, while ITD and ILD summarize the main interaural timing and level cues. These metrics are examined not only as global averages, but also across frequency bands and spatial regions. The main research question is whether the head-and-torso model reduces the simulation error globally, or only in specific frequency bands and spatial regions.

II. Data and Methodologies

A. SONICOM dataset

The analysis was carried out using subjects from the SONICOM HRTF dataset [6]. The dataset provides measured head-related transfer functions (HRTFs), headphone transfer functions, 3D scans of the ears, head and torso, and depth-image data acquired around each subject. This makes SONICOM particularly suitable for this work, since it allows measured HRTFs, numerical simulations and anatomical meshes to be analysed within the same database.

The HRTF data are distributed in SOFA format, i.e. the Spatially Oriented Format for Acoustics [8]. A SOFA file stores the impulse responses together with the corresponding spatial metadata, such as source positions, receiver information, sampling rate and listener-related descriptors. In this way, each HRIR can be directly associated with its azimuth and elevation, which is essential for comparing measured and simulated datasets on a common spatial grid.

[1] https://github.com/Any2HRTF/Mesh2HRTF
[2] https://github.com/Katarina-Poole/Spatial-Audio-Metrics

For each subject, SONICOM provides different measured HRIR versions, including raw, windowed, free-field compensated, and free-field compensated minimum-phase data. Versions with ITD removed are also available. In this work, the measured reference was the free-field compensated 48 kHz SOFA file, since this version reduces the influence of the measurement loudspeaker response and is therefore more consistent with comparisons against numerical simulations.

The SONICOM spatial grid contains 793 released source positions per subject, covering the full azimuth range and elevations from -45° to 90°. The angular convention adopted in this work is 0° for the frontal direction, $+90^\circ$ for the listener's left side, -90° for the listener's right side, and $\pm 180^\circ$ for the rear direction. Only directions belonging to the actual SONICOM grid were used in the analysis.

In addition to the measured HRTFs, the extended SONICOM release includes synthetic HRTFs generated with Mesh2HRTF and several mesh variants for each processed subject. These include preprocessed meshes, plugged meshes, and left/right graded meshes. The plugged meshes close the ear canal entrance, while the graded meshes preserve higher resolution near the active ear and reduce mesh density in less critical regions. For the present work, the SONICOM mesh resources were used as the starting point for the head-only simulation, while a modified head-and-torso mesh was used to evaluate the effect of adding torso geometry to the numerical HRTF simulation.

B. *Head-only and head-and-torso mesh preparation*

The head-only simulations required no meshing on our side. For each analysed subject, the extended SONICOM release already provides two graded surface meshes, one refined around the left ear and one around the right [7]; since Mesh2HRTF computes the transfer function for a single ear at a time, these were taken directly as the head geometry. The grading keeps small triangles around the active ear, where the short-wavelength scattering responsible for the high-frequency spectral cues takes place, and coarsens the mesh away from it, keeping the element count—and thus solver memory and runtime—manageable. These meshes derive from the watertight head scans distributed with the dataset, reconstructed from a point cloud sampled at 0.5 mm [6], and were used as-is in the simulation stage of Sec. II-C.

The head-and-torso geometry, by contrast, is not distributed in simulation-ready form and was prepared by us from the raw head-and-torso scan of subject P067 [6]. This scan reproduces the body together with the subject's clothing and the typical artefacts of structured-light scanning (holes, isolated fragments, non-manifold edges, and self-intersecting regions). Since the boundary element method requires a rigid, acoustically hard, watertight surface, the clothing was removed and the underlying torso and shoulder surface reconstructed and smoothed; the mesh was then repaired by filling holes, deleting loose and unreferenced geometry, resolving non-manifold and self-intersecting regions, and merging head and torso into a single closed manifold. Cleaning and repair were carried out in Blender and MeshLab, with removal of unreferenced vertices as the final step. The watertight surface was then graded once per ear, using the same criterion as the released head meshes so that the ear regions kept the required resolution [4], and passed to the same simulation and SOFA-export stage as the head-only geometry.

C. *Mesh2HRTF simulations and SOFA compatibility*

The head-only and head-and-torso HRTFs were simulated using Mesh2HRTF, an open-source framework for the numerical calculation of HRTFs based on the boundary element method. The simulations were exported as SOFA files and evaluated on the same SONICOM spatial grid used for the measured data. This allowed the simulated HRIRs to be directly compared with the measured SONICOM reference and with the official SONICOM simulated HRTFs.

These simulations are computationally demanding, with memory rather than processing speed as the binding constraint [11]: the high-frequency steps can require on the order of 20–30 GB each, so a multi-core machine with at least 64 GB of RAM was used, limiting how many frequencies could be solved concurrently. A complete per-ear run over the full frequency range takes several hours [10], with the head-and-torso geometry being the heaviest.

Before performing quantitative comparisons, a technical compatibility check was applied to all SOFA files. The check verified the SOFA convention, the shape of the HRIR matrices, the shape and ordering of the source-position arrays, the sampling rate, the number of spatial directions, and the length of the impulse responses. This step was necessary to avoid comparing datasets with inconsistent spatial grids, different sampling rates, incompatible SOFA conventions, or non-matching source directions.

The source-position validation was particularly important because all error metrics require that each HRIR in one dataset corresponds to the same physical direction in the other datasets. The adopted angular convention was the SONICOM/SOFA convention. The analysis was restricted to the actual SONICOM elevation values,

$$-45^\circ, -30^\circ, -20^\circ, -10^\circ, 0^\circ, 10^\circ, 20^\circ, 30^\circ, 45^\circ, 60^\circ, 75^\circ, 90^\circ.$$

Directions not present in the original grid were not introduced, in order to avoid nearest-neighbour ambiguities. After this validation, all datasets were considered compatible for direction-wise spectral and binaural metric comparisons.

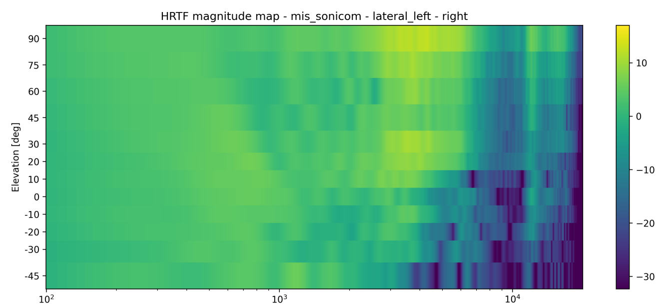

Fig. 1: Example of measured SONICOM HRTF magnitude map for subject P143. The map shows the right-ear magnitude response on the selected lateral-left plane as a function of frequency and elevation.

D. *Evaluation metrics and spatial analysis*

The evaluation was performed at two complementary levels. First, a descriptive HRTF analysis was carried out by inspecting HRIRs in the time domain, HRTF magnitude spectra in the frequency domain, and frequency–direction error maps. These maps were used to visualize how the difference between datasets varied with frequency, azimuth, elevation and ear. In particular, they allowed the identification of frequency bands and spatial regions where the simulated HRTFs were closer to, or farther from, the measured SONICOM reference.

The descriptive analysis considered three main comparisons: our Mesh2HRTF simulation against the measured SONICOM HRTF, the official SONICOM simulation against the measured SONICOM HRTF, and our Mesh2HRTF simulation against the official SONICOM simulation. The map-based analysis was used as a diagnostic tool rather than as a single performance indicator, since visual maps can reveal localized spectral discrepancies that may be hidden by global averages.

After this exploratory stage, compact quantitative metrics were computed for each matched source direction. The first metric was the mean absolute spectral error, defined from the difference between the HRTF magnitudes in dB. For two datasets A and B, the direction- and ear-dependent spectral error was computed as

$$E_{\mathrm{mag}}(\Omega, e) = \langle |H_{A,\mathrm{dB}}(f, \Omega, e) - H_{B,\mathrm{dB}}(f, \Omega, e)| \rangle_f \,, \qquad (1)$$

where the average is taken over the selected frequency range. This metric was used to summarize the overall spectral agreement between simulated and measured HRTFs.

In addition, interaural time difference (ITD), interaural level difference (ILD), and log-spectral distortion (LSD) were used as spatial-audio metrics. ITD quantifies the timing difference between the left and right HRIRs and is mainly related to lateralization. ILD measures the interaural level difference and is especially relevant for lateral and high-frequency cues. LSD evaluates the mismatch between log-magnitude spectra and is therefore sensitive to spectral peaks, notches and fine HRTF shape differences. ITD and ILD were estimated using the Spatial Audio Metrics framework, while LSD was computed from the frequency-domain HRTF magnitudes.

For each comparison, the metrics were first computed direction by direction and then averaged over selected spatial regions. The considered regions were the full analysed grid, the horizontal plane, the median plane, the frontal and rear median planes, and the left and right lateral planes. This regional analysis was necessary because a global average may hide localized improvements or degradations. In particular, the head-and-torso simulation was expected to affect lateral and low-elevation directions more strongly than frontal directions, due to shoulder and torso diffraction, reflection and shadowing.

The interpretation of the metrics was kept separated because they describe different acoustic properties. A simulation may improve ITD or ILD while still worsening LSD, meaning that the binaural timing or level cues can become more accurate even if the detailed spectral shape is less consistent with the measured HRTF. For this reason, the final evaluation combines global averages, frequency-band results and spatial-region analysis instead of relying on a single scalar metric.

E. Analysis protocol

The evaluation was organised in two stages. In the first stage, the metrics described above were used as a benchmark of the head-only Mesh2HRTF simulation pipeline. The purpose of this benchmark was to compare our simulation with both the measured SONICOM HRTF and the official SONICOM simulated HRTF, in order to assess whether our numerical pipeline produced results within a reasonable range with respect to the reference dataset. This stage considered the available frequency range of the datasets and was used to identify the main spectral and spatial discrepancies between measurement, official simulation and our head-only simulation.

In the second stage, the analysis focused on the specific effect of adding torso geometry. For this purpose, the head-only and head-and-torso simulations of the same subject were both compared against the measured SONICOM HRTF, and the comparison was restricted to frequencies up to 10 kHz. This upper limit is primarily dictated by the head-and-torso mesh itself. In a boundary element simulation, the highest frequency that can be reproduced reliably is set by the size of the mesh elements: roughly six elements are required per acoustic wavelength to ensure numerical accuracy [12], [4], [11], so the largest triangle anywhere on the surface fixes the upper valid frequency,

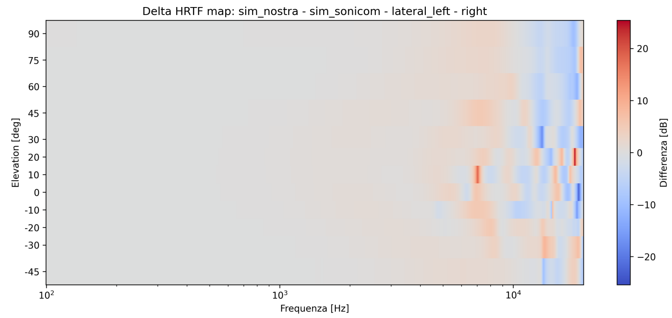

Fig. 2: Signed HRTF spectral difference map between our head-only Mesh2HRTF simulation of subject P0143 and the official SONICOM simulation on the selected lateral-left plane, right ear. The map shows our simulation minus the SONICOM simulation. Red regions indicate higher magnitude in our simulation, while blue regions indicate lower magnitude.

approximately $f_{\max} \approx c/(6\,L_{\max})$, where c is the speed of sound and $L_{\max}$ the longest element edge. On the graded head meshes this ceiling is high, because the elements are kept small around the active ear. The torso, however, is a large and smooth surface that would require an extremely large number of fine triangles to be meshed at the same resolution; to keep the problem tractable it is necessarily meshed with comparatively large elements. It is therefore the largest of these torso triangles that sets the valid-bandwidth ceiling of the head-and-torso simulation, which is consequently lower than that of the head-only model. Above this ceiling the simulated high-frequency content of the head-and-torso geometry is no longer numerically trustworthy, and restricting the comparison to 10 kHz keeps the evaluation within the band where the head-and-torso simulation is reliable.

This restriction is also desirable on physical grounds. At very high frequencies the HRTF is dominated by narrow spectral notches and by fine geometrical details of the pinna [1], [13], the receiver position and the head–torso junction, so that small geometrical or alignment differences can produce large spectral deviations that would dominate a global error metric without reflecting the actual contribution of the torso. The 10 kHz limit is therefore not intended to discard high-frequency information as such, but to separate the broad-band benchmark of the head-only pipeline from the torso-specific validation. Since torso and shoulder effects—diffraction, reflection and shadowing—are expected to introduce relatively broad spectral patterns at low and mid frequencies, and mainly in lateral and low-elevation directions [14], [9], the restricted-band analysis is used to test whether the head-and-torso model brings physically meaningful improvements precisely in the frequency and spatial regions where torso-related effects are most stable. The results are then interpreted using both global averages and frequency-/region-dependent trends, avoiding conclusions based on a single scalar error value.

III. Results

A. Spectral-error analysis

The spectral comparison was based on the mean absolute spectral error between the simulated subject P067 HRTFs and the measured SONICOM P067 HRTF. Before the comparison, the measured HRTF was aligned in level with the mean of the head-only and head-and-torso simulations using a broadband normalization over 500–6000

TABLE I: Mean spectral error against measured SONICOM HRTF for different frequency bands. Positive ΔE indicates an improvement of the head-and-torso simulation.

Band [Hz]	Head-only	Head+torso	ΔE
100–10000	5.161	5.315	-0.154
100–500	2.721	2.204	0.517
500–1500	2.593	2.415	0.178
1500–3000	1.693	1.452	0.241
3000–6000	2.473	2.194	0.279
6000–10000	9.353	10.129	-0.776

TABLE II: Mean spectral error against measured SONICOM HRTF in the lateral plane. Positive ΔE indicates an improvement of the head-and-torso simulation.

Band [Hz]	Head-only	Head+torso	ΔE
100–10000	5.557	4.994	0.563
100–500	2.825	2.228	0.597
500–1500	2.311	2.515	-0.204
1500–3000	1.729	1.309	0.421
3000–6000	2.908	2.039	0.869
6000–10000	10.052	9.476	0.576

Hz. For each table, the improvement is defined as

$$\Delta E = E_{\mathrm{head-only}} - E_{\mathrm{head+torso}}, \tag{2}$$

so that positive values indicate that the head-and-torso simulation is closer to the measured reference.

Table I shows that the effect of adding the torso is strongly frequency-dependent. From 100 Hz to 6 kHz, the head-and-torso model consistently reduces the spectral error with respect to the head-only model. The largest improvement is found in the 100–500 Hz band, where the error decreases from 2.721 dB to 2.204 dB. Smaller but consistent improvements are also observed up to 6 kHz. In contrast, the 6000–10000 Hz band worsens from 9.353 dB to 10.129 dB. Consequently, the overall 100–10000 Hz selected-direction average remains slightly better for the head-only simulation, despite the improvements observed at low and mid frequencies.

The spatial analysis clarifies that the full-band degradation is not uniform across directions. Over the selected directions, the head-and-torso model slightly worsens the 100–10000 Hz average, from 5.161 dB to 5.315 dB. The same trend is observed in the horizontal and median planes, where the error increases from 5.114 dB to 5.507 dB and from 4.696 dB to 5.366 dB, respectively. However, the lateral plane shows the opposite behaviour: the error decreases from 5.557 dB to 4.994 dB. This is consistent with the expected role of shoulder and torso diffraction, reflection, and shadowing in lateral directions.

The median-plane result is useful for interpreting the global average. In this plane, the head-and-torso model improves the error below 3 kHz, but the 6000–10000 Hz band worsens markedly, from 8.556 dB to 10.212 dB. This high-frequency degradation dominates the full-band median-plane average and contributes substantially to the global worsening of the head-and-torso simulation. Therefore, the global average should not be interpreted as evidence that the torso has no useful effect, but rather as the result of a compensation between low-/mid-frequency improvements and larger high-frequency discrepancies in non-lateral regions.

For this reason, the lateral plane was analysed separately in frequency bands. Table II shows that the head-and-torso model improves the lateral-plane error over almost the entire frequency range. The strongest improvement is observed in the 3000–6000 Hz band, where the error decreases from 2.908 dB to 2.039 dB. Unlike the selected-direction average, the lateral plane also improves in the 6000–10000 Hz band, suggesting that the high-frequency degradation mainly originates from non-lateral directions.

A further analysis by elevation shows that the lateral-plane improvement is not uniform, but is mainly driven by low-elevation lateral directions. Table III reports the full-band and high-frequency improvement for selected lateral elevations. The largest full-band improvements occur at negative elevations, especially between -45° and -20°. In these directions, the head-and-torso simulation also improves the 6000–10000 Hz band, whereas around the horizontal plane the improvement becomes small or negative. This indicates that the high-frequency degradation observed in the selected-direction average is not a general property of the torso simulation, but is spatially localized mainly outside low-elevation lateral directions.

TABLE III: Spectral-error improvement for selected lateral elevations. Positive ΔE indicates that the head-and-torso simulation is closer to the measured HRTF.

Lateral elevation	ΔE 100–10000 Hz	ΔE 6000–10000 Hz
-45°	3.75	4.28
-30°	2.25	3.59
-20°	2.00	3.79
-10°	-0.70	-2.17
$+0^\circ$	0.06	-0.39

Overall, the spectral-error results show that the torso contribution is frequency-, direction-, and elevation-dependent. The head-and-torso simulation improves the agreement below 6 kHz and clearly improves the lateral plane, with the strongest benefit occurring for low-elevation lateral directions. However, these improvements are not sufficient to reduce the overall selected-direction average, because non-lateral high-frequency discrepancies, especially in the median plane and around near-horizontal elevations, have a strong influence on the full-band.

B. SAM analysis

The SAM analysis was used to complement the spectral-error results with ITD, ILD, and LSD metrics. Table IV reports the main results for the full SONICOM evaluation grid and for the lateral plane. Over the full grid, the head-and-torso simulation does not improve the agreement with the measured HRTF: ITD, ILD, and LSD are all worse than for the head-only simulation. The official SONICOM simulation gives the best global ITD and LSD values, while the head-only model gives the best global ILD value.

The most relevant SAM result is the LSD trend, because LSD is directly related to log-spectral differences and is therefore consistent with the spectral-error analysis. Globally, the head-and-torso model worsens the LSD, from 7.23 dB to 7.68 dB. In the lateral plane, however, the LSD improves from 8.22 dB to 7.66 dB, confirming that the torso improves the lateral spectral shape while remaining penalized by discrepancies in other regions. The official SONICOM simulation remains the best solution in LSD, both globally and laterally.

TABLE IV: SAM metrics against measured SONICOM HRTFs over 100–10000 Hz. Lower values indicate better agreement with the measured reference.

Region	Metric	Head-only	Head+torso	SONICOM sim.
Global	ITD [μs]	21.02	23.64	20.60
Global	ILD [dB]	1.45	1.70	1.65
Global	LSD [dB]	7.23	7.68	6.47
Lateral	ITD [μs]	38.83	25.57	39.77
Lateral	ILD [dB]	1.80	2.31	1.98
Lateral	LSD [dB]	8.22	7.66	7.45

The ITD results also show a lateral reduction, from 38.83 μs to 25.57 μs, but this should be interpreted as a secondary indication: the difference is on the order of a few tens of microseconds and the ITD estimator operates on HRIRs band-limited to 10 kHz. Conversely, ILD does not improve. The ILD error increases both globally and in the lateral plane, indicating that the improved lateral spectral shape does not translate into a more accurate interaural level balance.

Overall, the SAM metrics support the same conclusion as the spectral-error analysis. The head-and-torso simulation is not globally superior to the head-only model. Its most robust benefit is a spatially selective improvement of the lateral spectral agreement, with a secondary indication of improved lateral temporal behaviour, while ILD remains a limitation.

IV. Conclusion

This work evaluated head-only and head-and-torso HRTF simulations against measured SONICOM data. The results indicate that the effect of adding the torso is frequency-, direction-, and elevation-dependent, rather than consistently beneficial over the whole analysed grid. In the spectral-error analysis, the head-and-torso simulation shows lower error than the head-only model below 6 kHz and, most noticeably, in lateral directions. A more detailed elevation analysis suggests that this lateral behaviour is mainly associated with low-elevation sources, especially between approximately -45° and -20°, where torso and shoulder effects are expected to be more relevant. However, the global 100–10000 Hz average remains slightly worse than the head-only simulation, mainly due to larger discrepancies in the 6–10 kHz range and in non-lateral regions, especially the median plane.

The SAM metrics lead to a consistent interpretation. The LSD results, which are directly related to log-spectral differences, follow the same trend as the spectral-error analysis: the head-and-torso model gives a higher global LSD, but a lower LSD in the lateral plane. This suggests that the torso can improve the lateral spectral agreement, while not reducing the overall spectral mismatch. The lateral ITD error is also reduced when the torso is included, which may indicate a more consistent temporal behaviour in lateral directions; however, this result should be interpreted cautiously because the absolute differences are on the order of a few tens of microseconds. Conversely, ILD does not improve, indicating that the added torso does not reproduce the measured interaural level balance more accurately in this configuration.

Overall, the head-and-torso simulation cannot be considered globally superior to the head-only model. Its main effect appears to be spatially selective, with the most consistent benefit observed in the lateral spectral agreement, particularly for low-elevation lateral directions. These findings suggest that adding a torso can be useful for representing specific spatial features of the HRTF, but that accurate subject-specific torso geometry, head–torso alignment, and mesh modelling are still required before a robust global improvement over head-only HRTF simulation can be expected.

Finally, the present analysis is also constrained by computational and meshing limitations. Including the torso substantially increases the size and complexity of the boundary element problem, so the torso surface had to be meshed more coarsely than the ear and head regions in order to keep the simulations computationally feasible. This limits the reliable frequency range of the head-and-torso model and makes the results above approximately 10 kHz unsuitable for robust validation. Therefore, the observed high-frequency discrepancies should be interpreted not only as acoustic differences between geometries, but also as partly influenced by the practical trade-off between anatomical detail, mesh resolution, and available computational resources. Future work should investigate whether higher-resolution subject-specific torso meshes and larger computational resources can reduce these residual discrepancies.

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